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Q1. 1) Prokaryotic cells have no nucleus - their DNA lies free in the cytoplasm in a nucleoid; eukaryotic cells have a membrane-bound nucleus. 2) Prokaryotes lack membrane-bound organelles such as mitochondria and ER; eukaryotes have them. 3) Prokaryotes have smaller 70S ribosomes; eukaryotes have 80S ribosomes.

Q2. Mitochondria are the site of aerobic respiration (Krebs cycle and electron transport chain), producing ATP which the cell uses as energy. Muscle cells contract continuously and need a lot of ATP, so they contain many mitochondria to release enough energy.

Q3. Enzymes are biological catalysts. Each enzyme has an active site with a specific shape complementary to its substrate (lock and key; in the induced-fit model the active site moulds around the substrate). The substrate binds to form an enzyme-substrate complex, which lowers the activation energy so the reaction happens faster; the products are released and the enzyme is unchanged.
Temperature: as temperature rises, molecules gain kinetic energy and collide more often, so the rate increases up to the optimum (about 37 C for human enzymes). Above the optimum, bonds holding the tertiary structure break, the active site changes shape and the enzyme is denatured, so activity falls rapidly.
pH: each enzyme has an optimum pH (pepsin about 2, amylase about 7). Moving away from it changes the charges on amino acids, breaking ionic and hydrogen bonds, which alters the active site so the substrate no longer fits.